40.4: Energy requirements are related to animal size, activity, and environment

One of the unifying themes of biology, introduced in Concept 1.1, is that life requires energy transfer and transformation. Like other organisms, animals use chemical energy for growth, repair, activity, and reproduction. The overall flow and transformation of energy in an animal—its ****bioenergetics—determines nutritional needs and is related to the animal's size, activity, and environment.

Energy Allocation and Use

Organisms can be classified by how they obtain chemical energy. Most *autotrophs*, such as plants, harness light energy to build energy-rich organic molecules and then use those molecules for fuel. Most *heterotrophs*, such as animals, obtain their chemical energy from food, which contains organic molecules synthesized by other organisms.

Animals use chemical energy harvested from the food they eat to fuel metabolism and activity. Food is digested by enzymatic hydrolysis (see Figure 5.2b), and nutrients are absorbed by body cells. The ATP (adenosine triphosphate) produced by cellular respiration and fermentation powers cellular work, enabling cells, organs, and organ systems to perform the functions that keep an animal alive. Other uses of energy in the form of ATP include biosynthesis, which is needed for body growth and repair, synthesis of storage material such as fat, and production of gametes. The production and use of ATP generate heat, which the animal eventually gives off to its surroundings **(Figure 40.18)**.

Figure 40.18

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Bioenergetics of an animal: an overview.
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Flow of processes in the animal body following the intake of organic molecules in food. Processes outlined include digestion and absorption, cellular respiration, biosynthesis, and cellular work.

Flow of processes in the animal body following the intake of organic molecules in food. Processes outlined include digestion and absorption, cellular respiration, biosynthesis, and cellular work. The animal body takes in organic molecules in food. During digestion and absorption, some heat is lost to the environment, and body cells uptake nutrient molecules from the food. Some of the energy from food is lost to the environment as nitrogenous waste, and some of the nutrient molecules are used in cellular respiration (more

heat is lost to the environment) or for their carbon skeletons. The carbon skeletons are used in biosynthesis, which includes growth, storage and reproduction (and some heat is lost to the environment during biosynthesis). Cellular respiration provides ATP for cellular work (some of which includes biosynthesis). During cellular work more heat is lost to the environment.

MAKE CONNECTIONS*Use the idea of energy coupling to explain why heat is produced in the absorption of nutrients, in cellular respiration, and in the synthesis of biopolymers (see*Concept 8.3*).**

*For suggested answer, see*Appendix A*.**

Quantifying Energy Use

How much of the total energy an animal obtains from food does it need just to stay alive? How much energy must be expended to walk, run, swim, or fly from one place to another? What fraction of the energy intake is used for reproduction? Physiologists answer such questions by measuring the rate at which an animal uses chemical energy and how this rate changes in different circumstances.

The sum of all the energy an animal uses in a given time interval is called its **metabolic rate**. Energy is measured in joules (J) or in calories (cal) and kilocalories (kcal). A kilocalorie equals 1,000 calories, or 4,184 joules. (The unit Calorie, with a capital C, as used by many nutritionists, is actually a kilocalorie.)

Metabolic rate can be determined in several ways. Because nearly all of the chemical energy used in cellular respiration eventually appears as heat, metabolic rate can be measured by monitoring an animal's rate of heat loss. For this approach, researchers use a calorimeter, which is a closed, insulated chamber equipped with a device that records the heat an animal gives off to its environment. Metabolic rate can also be determined from the amount of oxygen consumed or carbon dioxide produced by an animal's cellular respiration **(Figure 40.19)**. To calculate metabolic rate over longer periods, researchers record the rate of food consumption, the energy content of the food (about 4.5–5 kcal per gram of protein or carbohydrate and about 9 kcal per gram of fat), and the chemical energy lost in waste products (feces and urine or other nitrogenous wastes).

Figure 40.19

Measuring the rate of oxygen consumption by a swimming shark.

A juvenile hammerhead shark swims in a tank while a researcher makes observations.

A researcher monitors the decrease in oxygen level over time in the recirculating water of a juvenile hammerhead's tank.

Minimum Metabolic Rate and Thermoregulation

Animals must maintain a minimum metabolic rate for basic functions such as cell maintenance, breathing, and circulation. Researchers measure this minimum metabolic rate differently for endotherms and ectotherms. The minimum metabolic rate of a nongrowing endotherm that is at rest, has an empty stomach, and is not experiencing stress is called the **basal metabolic rate (BMR)**. BMR is measured under a "comfortable" temperature range—a range that requires only the minimum generation or shedding of heat. The minimum metabolic rate of ectotherms is determined at a specific temperature because changes in the environmental temperature alter body temperature and therefore metabolic rate. The metabolic rate of a fasting, nonstressed ectotherm at rest at a particular temperature is called its **standard metabolic rate (SMR)**.

Comparisons of minimum metabolic rates reveal the different energy costs of endothermy and ectothermy. The BMR for humans averages 1,600–1,800 kcal per day for adult males and 1,300–1,500 kcal per day for adult females. These BMRs are about equivalent to the rate of energy use by a 75-watt lightbulb. In contrast, the SMR of an American alligator is only about 60 kcal per day at 20°C (68°F). As this represents less than 120 the energy used by a comparably sized adult human, it is clear that ectothermy has a markedly lower energetic requirement than endothermy.

Influences on Metabolic Rate

Metabolic rate is affected by many factors other than an animal being an endotherm or an ectotherm. Some key factors are age, sex, size, activity, temperature, and nutrition. Here we'll examine the effects of size and activity.

Size and Metabolic Rate

Larger animals have more body mass and therefore require more chemical energy. Remarkably, the relationship between overall metabolic rate and body mass is constant across a wide range of sizes and forms, as illustrated for various mammals in **Figure 40.20a**. In fact, for even more varied organisms ranging in size from bacteria to blue whales, metabolic rate remains roughly proportional to body mass to the three-quarter power (m3/4). Scientists are still researching the basis of this relationship, which applies to ectotherms as well as endotherms.

Figure 40.20

The relationship of metabolic rate to body size.

Two graphs are given.

First. Graph of basal metabolic rate (BMR) by body mass for various mammals of increasing size (shrew to elephant). BMR increases steadily with body size. The x-axis is body mass in kilograms (log scale), ranging from 10 to the negative 3 power to 10 to the third power. The y-axis is BMR (liters of O 2 per hour, log scale), ranging from 10 to the negative 2 power to 10 to the third power. BMR increases with body mass from the shrew to the elephant. Second. Graph of basal metabolic rate per kilogram of body mass to body size for various animals. The x-axis is body mass (kilograms log scale), ranging from 10 to the negative 3 power to 10 to the third power; and the y-axis is BMR (liters O 2 per hour) per kilogram, ranging from 0 to 8. There is a sharp decrease in BMR per kilogram with increasing body mass. The decline is steepest between the shrew and the harvest mouse. The line then flattens as body mass increases and BMR per kilogram approaches 0.

INTERPRET THE DATA *Based on the graph in (a), one observer suggests that a group of* 100 *ground squirrels has the same basal metabolic rate as* 1 *dog. A second observer looking at the graph disagrees. Who is correct and why?*

The relationship of metabolic rate to size profoundly affects energy consumption by body cells and tissues. As shown in **Figure 40.20b**, the energy it takes to maintain each gram of body mass is inversely related to body size. Each gram of a mouse, for instance, requires about 20 times as many calories as a gram of an elephant, even though the whole elephant uses far more calories than the whole mouse. The smaller animal's higher metabolic rate per gram demands a higher rate of oxygen delivery. To meet this demand, the smaller animal must have a higher breathing rate, blood volume (relative to its size), and heart rate.

Thinking about body size in bioenergetic terms reveals how trade-offs shape the evolution of body plans. As body size decreases, each gram of tissue increases in energy cost. As body size increases, energy costs per gram of tissue decrease, but an ever-larger fraction of body tissue is required for exchange, support, and locomotion.

Activity and Metabolic Rate

For both ectotherms and endotherms, activity greatly affects metabolic rate. Even a person reading quietly at a desk or an insect twitching its wings consumes energy beyond the BMR or SMR. Maximum metabolic rates (the highest rates of ATP use) occur during peak activity, such as lifting a heavy object, sprinting, or swimming at high speed. In general, the maximum metabolic rate an animal can sustain is inversely related to the duration of activity.

For most terrestrial animals, the average daily rate of energy consumption is two to four times BMR (for endotherms) or SMR (for ectotherms). Humans in most developed countries have an unusually low average daily metabolic rate of about 1.5 times BMR—an indication of a relatively sedentary lifestyle.

The fraction of an animal's energy "budget" that is devoted to activity depends on many factors, including its environment, behavior, size, and thermoregulation. In the **Scientific Skills Exercise**, you'll interpret data

on the annual energy budgets of three terrestrial vertebrates.

Scientific Skills Exercise

Interpreting Pie Charts

AP Science Practices 2, 4, 6

How Do Energy Budgets Differ for Three Terrestrial Vertebrates? To explore bioenergetics in animal bodies, let's consider typical annual energy budgets for three terrestrial vertebrates that vary in size and thermoregulatory strategy: a 4-kg male Adélie penguin, a 25-g(0.025-kg) female deer mouse, and a 4-kg female ball python. The penguin is well-insulated against his Antarctic environment but must expend energy in swimming to catch food, incubating eggs laid by his partner, and bringing food to his chicks. The tiny deer mouse lives in a temperate environment where food may be readily available, but her small size causes rapid loss of body heat. Unlike the penguin and mouse, the python is ectothermic and keeps growing throughout her life. She produces eggs but does not incubate them. In this exercise, we'll compare the energy expenditures of these animals for five important functions: basal (standard) metabolism, reproduction, thermoregulation, activity, and growth.

How the Data Were Obtained Energy budgets were calculated for each of the animals based on measurements from field and laboratory studies.

Data from the Experiments Pie charts are a good way to compare *relative* differences in a set of variables. In the pie charts here, the sizes of the wedges represent the relative annual energy expenditures for the functions shown in the key. The total annual expenditure for each animal is given below its pie chart.

Three pie charts summarizing energy budgets for Adelie penguin, deer mouse and ball python.

Data from M. A. Chappell et al., Energetics of foraging in breeding Adélie penguins, *Ecology* 74:2450–2461 (1993); M. A. Chappell et al., Voluntary running in deer mice: speed, distance, energy costs, and temperature effects, *Journal of Experimental Biology* 207:3839–3854 (2004); T. M. Ellis and M. A. Chappell, Metabolism, temperature relations, maternal behavior, and reproductive energetics in the ball python (*Python regius*), *Journal of Comparative Physiology B* 157:393–402 (1987).

The charts are as follows.

Adelie Penguin. The total energy budget for the 4-kilogram male Adelie penguin is 340,000 kilocalories per year; the largest portions of his pie chart go to activity and basal metabolism.

Deer mouse. The total energy budget for the 0.025 kilogram female deer mouse is 4,000 kilocalories per year; the largest portions of the pie chart go to thermoregulation, followed by roughly equal proportions to the other categories in the budget.

Ball python. The total energy budget for the 4 kilogram female ball python is 8,000 kilocalories per year; the largest portions of the pie chart go to activity and basal metabolism.

INTERPRET THE DATA

- You can estimate the contribution of each wedge in a pie chart by remembering that the entire circle represents 100%, half is 50%, and so on. What percent of the mouse's energy budget goes to basal metabolism? What percent of the penguin's budget is for activity?
- Without considering the sizes of the wedges, how do the three pie charts differ in which functions they include? Explain these differences.
- Does the penguin or the mouse expend a greater proportion of its energy budget on thermoregulation? Why?
- Now look at the *total* annual energy expenditures for each animal. How much more energy does the penguin expend each year compared to the similarly sized python?
- Which animal expends the most kilocalories per year on thermoregulation?
- If you monitored energy allocation in the penguin for just a few months instead of an entire year, you might find the growth category to be a significant part of the pie chart. Given that adult penguins don't grow from year to year, how would you explain this finding?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Torpor and Energy Conservation

Despite their many adaptations for homeostasis, animals may encounter conditions that severely challenge their abilities to balance their heat, energy, and materials budgets. For example, at certain times of the day or year, their surroundings may be extremely hot or cold, or food may be unavailable. A major adaptation that enables animals to save energy in the face of such difficult conditions is **torpor**, a physiological state of decreased activity and metabolism.

Many birds and small mammals exhibit a daily torpor that is well adapted to feeding patterns. For instance, some bats feed at night and go into torpor in daylight. Similarly, chickadees and hummingbirds, which feed during the day, often go into torpor on cold nights.

All endotherms that exhibit daily torpor are relatively small; when active, they have high metabolic rates and thus very high rates of energy consumption. The changes in body temperature, and thus the energy savings,

are often considerable: the body temperature of chickadees drops as much as 10°C (18°F) at night, and the core body temperature of a hummingbird can fall 25°C (45°F) or more.

Hibernation is long-term torpor that is an adaptation to winter cold and food scarcity. When a mammal enters hibernation, its body temperature declines as its body's thermostat is turned down **(Figure 40.21)**. Some hibernating mammals cool to as low as 1–2°C (34–36°F), and at least one, the Arctic ground squirrel (*Spermophilus parryii*), can enter a supercooled (unfrozen) state in which its body temperature dips below 0°C (32°F). Periodically, perhaps every two weeks or so, hibernating animals undergo arousal, raising their body temperature and becoming active briefly before resuming hibernation. Metabolic rates during hibernation can be 20 times lower than if the animal attempted to maintain normal body temperatures of 36–38°C (97–100°F). As a result, hibernators such as the ground squirrel can survive through the winter on limited supplies of energy stored in the body tissues or as food cached in a burrow. Similarly, the slow metabolism and inactivity of *estivation*, or summer torpor, enable animals to survive long periods of high temperatures and scarce water.

Figure 40.21

A hazel dormouse (*Muscardinus avellanarius*) hibernating.

A dormouse is curled up in a nest.

What happens to the circadian rhythm in hibernating animals? In the past, researchers reported detecting daily biological rhythms in hibernating animals. However, in some cases the animals were probably in a state of torpor from which they could readily arouse, rather than "deep" hibernation. More recently, a group of researchers in France addressed this question in a different way, examining the machinery of the biological clock rather than the rhythms it controls **(Figure 40.22)**. Working with the European hamster, they found that molecular components of the clock stopped oscillating during hibernation. These findings support the hypothesis that the circadian clock ceases operation during hibernation, at least in this species.

Figure 40.22

Inquiry: What happens to the circadian clock during hibernation?

Experiment

To determine whether the 24-hour biological clock continues to run during hibernation, Paul Pévet and colleagues at the University of Louis Pasteur in Strasbourg, France, studied molecular components of the circadian clock in the European hamster (*Cricetus cricetus*). The researchers measured RNA levels for two

clock genes—Per2 and Bmal1—during normal activity (euthermia) and during hibernation in constant darkness. The RNA samples were obtained from the suprachiasmatic nuclei (SCN), a pair of structures in the mammalian brain that control circadian rhythms.

Results

Bar chart of relative RNA levels for two clock genes, Per2 and Bmal1.

The horizontal axis shows the categories euthermia and hibernation with a bar for day and night for each gene and each category. The vertical axis is relative RNA level (percent). For both genes, 100% relative RNA levels are only seen during euthermia, and not during hibernation. Per2 reaches this during the day and Bmal1 reaches this at night.

Conclusion

Hibernation disrupted circadian variation in the hamster's clock gene RNA levels. Further experiments demonstrated that this disruption was not simply due to the dark environment during hibernation, since for nonhibernating animals RNA levels during a darkened daytime were the same as in daylight. The researchers concluded that the biological clock stops running in hibernating European hamsters and, perhaps, in other hibernators as well.

Data from F. G. Revel et al., The circadian clock stops ticking during deep hibernation in the European hamster, *Proceedings of the National Academy of Sciences USA* 104:13816–13820 (2007).

WHAT IF? *Suppose you discovered a new hamster gene and found that the levels of RNA for this gene were constant during hibernation. What could you conclude about the day and night RNA levels for this gene during euthermia?*

*For suggested answer, see*Appendix A*.*

From tissue types to homeostasis, this chapter has focused on the whole animal. We also investigated how animals exchange materials with the environment and how size and activity affect metabolic rate. For much of the rest of this unit, we'll explore how specialized organs and organ systems enable animals to meet the basic challenges of life. In Unit 6, we investigated how plants meet the same challenges. **Figure 40.23**, on the next two pages, highlights some fundamental similarities and differences in the evolutionary adaptations of plants and animals. This figure is thus a review of Unit 6, an introduction to Unit 7, and, most importantly, an illustration of the connections that unify the myriad forms of life.

Figure 40.23

AP Make Connections: Life Challenges and Solutions in Plants and Animals

The figure compares three ways in which plants and animals respond to seven life challenges: nutritional mode, growth and regulation, environmental response, transport, reproduction, gas exchange, and absorption. The figure compares four ways in which plants and animals respond to seven life challenges: transport, reproduction, gas exchange, and absorption.

Multicellular organisms face a common set of challenges. Comparing solutions that have evolved in plants and animals reveals both unity (shared elements) and diversity (distinct features) across these two lineages.

Nutritional mode: All living things must obtain energy and carbon from the environment to grow, survive, and reproduce. Plants are autotrophs, obtaining their energy through photosynthesis and their carbon from inorganic sources, whereas animals are heterotrophs, obtaining their energy and carbon from food. Evolutionary adaptations in plants and animals support these different nutritional modes. The broad surface of many leaves enhances light capture for photosynthesis. When hunting, a bobcat relies on stealth, speed, and sharp claws. (See figure 36.2 and figure 41.16.)

Growth and regulation: the growth and physiology of both plants and animals are regulated by hormones. In plants, hormones may act in a local area or be transported in the body. They control growth patterns, flowering, fruit development, and more. In animals, hormones circulate throughout the body and act in specific target tissues, controlling homeostatic processes and developmental events such as molting. (See figure 39.5 and figure 45.12.)

Environmental response: all forms of life must detect and respond appropriately to conditions in their environment. Specialized organs sense environmental signals. For example, the floral head of the sunflower and an insect's eyes both contain photoreceptors that detect light. Environmental signals activate specific receptor proteins, triggering signal transduction pathways that initiate cellular responses coordinated by chemical and electrical communication. (See figure 39.19 and figure 50.15.)

Transport: all but the simplest multicellular organisms must transport nutrients and waste products between locations in the body. A system of tube-like vessels is the common evolutionary solution, while the mechanism of circulation varies. Plants harness solar energy to transport water, minerals, and sugars through specialized tubes. In animals, a pump (heart) moves circulatory fluid through vessels. (See figure 35.10 and figure 42.9.)

Reproduction: in sexual reproduction, specialized tissues and structures produce and exchange gametes. Offspring are generally supplied with nutritional stores that facilitate rapid growth and development. For example, seeds have stored food reserves that supply energy to the young seedling, while milk provides sustenance for juvenile mammals. (See figure 38.8 and figure 46.7.)

Gas exchange: the exchange of certain gases with the environment is essential for life. Respiration by plants and animals requires taking up oxygen (O_2) and releasing carbon dioxide (CO_2). In photosynthesis, net

exchange occurs in the opposite direction: CO_2 uptake and O_2 release. In both plants and animals, highly convoluted surfaces that increase the area available for gas exchange have involved, such as the spongy mesophyll of leaves and the alveoli of lungs. (See figure 35.18 and figure 42.24.)

Absorption: organisms need to absorb nutrients. The root hairs of plants and villi (projections) that line the intestines of vertebrates increase the surface area available for absorption. (See figure 28.4 and figure 41.13.)

The parts are as follows.

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MAKE CONNECTIONS *Compare the adaptations that enable plants and animals to respond to the challenges of living in hot and cold environments. See* ***Concept 39.4*** *and*Concept 40.3.*

*For suggested answer, see*Appendix A.*

BioFlix® Animation

Homeostasis: Regulating Blood Sugar

image

image

BioFlix® Animation

Water Transport in Plants

image

image

BioFlix® Animation

Gas Exchange

image

Play

image

Interview

Interview with George Bartholomew: Exploring connections between animal physiology and the environment

A photo of George Bartholomew.